Red Mountain Road Upgrade 2026
Statement of Work (SOW)

Bureau of Land Management (BLM)
Arcata Field Office

1.0 Background

The Bureau of Land Management (BLM) Arcata Field office requires work that will entail repairs, upgrades and reconstruction of road drainage sites along 12 miles of Red Mountain Road, a rural dirt and aggregate-surfaced road in Mendocino County, California near the community of Piercy (Maps 1 and 2). The road provides access to both public and private lands. The work is being done to replace aging infrastructure (e.g., culverts), comply with Clean Water Act requirements for reducing road-related erosion, and enhance safe vehicular access along the road. Specific actions include culvert replacements, rolling dip construction, aggregate base installation, road reshaping, and rock armoring of vulnerable fill slopes.

The work will conform to various specifications and Best Management Practices:

(a) Typical diagrams describing the various treatment types are provided as Appendix A of this Statement of Work (SOW).

(b) Best Management Practices for protection of water quality are provided in Appendix B of this SOW.

2.0 Scope

Road upgrading work shall be implemented along 12 miles of Red Mountain Road. The road is a rural, dirt and aggregate-surfaced road in a steep, mountainous setting. Specific treatment sites and materials quantities are listed in Tables 1 and 2, respectively. Broadly, the work includes:

- Site-specific upgrades at 12 sites along the road, treatments shall include:
  - Culvert replacements
  - Installation of rocked drivable rolling dips and critical dips (see Appendix A for diagrams)
  - Rock armoring of culvert fill slopes and inlet areas
  - Installation of rock base material on delineated road segments.
- Two additional sites are provided as <u>additive line items</u> for potential additional work.

Contractor is responsible for preparing the roadway for equipment access. This includes road brushing and road smoothing. Existing drainage features such as rolling dips shall not be disturbed unless approved by the Contracting Officer’s Representative (COR).

Some rolling dips may be abrupt and require modification prior to dump truck transit.

Prior to the start of work, the BLM will stake and/or flag the individual treatment sites.

3.0 Objectives

The objectives of the project are to:

- Comply with Clean Water Act requirements for appropriately functioning drainage infrastructure
- Provide for safe vehicle access

4.0 Requirements

The Contractor shall perform all site-specific treatments. The contractor shall be responsible for purchasing and providing all materials for the project, including necessary water for compaction. A limited quantity of rock is available at a small quarry site and is described in the Government Furnished Property section below and displayed in the attached Map 2.

Site Specific Treatments. The Contractor will implement specific treatments at 12 sites identified for upgrading. The Site-Specific Treatments Table 1 below provides the list of sites, treatments and the identified problem at the site as background.

Materials quantities for each site are listed in Table 2.

Installation methods and finished sites will conform to the typical drawings provided in Appendix A. Specific treatment types such as various rolling dip types, critical dips and culvert installations will adhere to the construction techniques provided in the typical drawings in this SOW. These include such measures as soils compaction, finished grades and mulching. The Contractor shall be responsible for reviewing and familiarizing themselves with the required treatment designs. Best Management Practices (BMP) for water quality protection are provided as Appendix B and are to be implemented in the appropriate circumstances as described for the specific BMP.

Each treatment site has the following minimum requirements:

- Contractor shall use angular road base material on all road and dip surfaces. Quantities are specified Table 2.
- Contractor shall use Class II 1.5” angular road base for all rolling dip, critical dip and road segments. Class II base will conform with 2023 CalTrans Standard Specifications.
- Rock will be applied to cover the entire constructed length and width of the individual dips.
- All rock will be compacted using a roller drum or other suitable compaction equipment. Tire-rolling of rock is not a suitable means of compaction.

- Approximately 3 tons of 3” angular ballast rock will be applied to the axis of each rolling dip. Critical dips do not require this. (EXHIBIT 3: PWA HANDBOOK SPECIFICATIONS)
- Culverts shall be set into prepared native material free of visible organic debris and rock particles >3”. Compaction shall occur in a minimum of 6” lifts. Water used to achieve compaction shall be supplied by the contractor.
- Contractor shall provide sufficient water to achieve compaction at all treatment sites.
- Contractor will apply straw mulch over all disturbed ground within 100 feet of a watercourse.
- Contractor shall utilize Aluminized, Type 2 corrugated metal pipes for all culvert sites. All culverts 42" or smaller shall be 12 gauge aluminized corrugated metal pipe. All culverts 48" and larger shall be 10 gauge aluminized corrugated metal pipe.
- Replaced culverts shall be transported and disposed of legally offsite.
- Fill slope armoring will use 6” to 24” rip rap materials and be installed to a minimum depth of 18”.
- Rolling dip and Critical Dip installations shall conform to specifications provided in Appendix A with emphasis on finished grades that provide for access for a variety of long wheel-based vehicles. Extensive excavation into the subgrade is expected to achieve these finished grades. Finished grades include not only approaches but dip outslope grades. Where possible, contractor shall salvage existing rock for re-application on to finished surfaces. Contractor shall demonstrate finished grades to COR upon completion.

Water Diversion:
If water is still running in the streams at the time of construction, the contractor shall divert water around the work area and back into the stream prior to working in the stream channel. Provide temporary bypass piping and cofferdams at the inlet.

Site Specific Treatments Table

Site #	Treatment
ALL	
2	1. Install critical dip on left side of #2 Xing. 2. Replace culvert (CMP) with 54" x 60' CMP set with outlet at the base of the outboard fill slope (OBF).
24	1. Replace existing pipe with new 48" x 40'L CMP set with inlet set ~ 6' - 8' closer to road surface and outlet at base of OBF. 2. Armor the OBF with 30 tons of 0.5' x 2' rock. 3. Cut inboard ditch (IBD) ~8' to right – mulch 4. Install 3 rolling dips up the right road. Connect to IBD. (May only be able to dip outer 1/2 of the road because the road is so steep). (Note: Critical dip omitted due to steep road and large pipe and a critical dip at nearby Site #25).
25	1. Replace existing pipe with new 24" x 50' CMP with outlet installed at the base of the OBF. 2. Reconstruct the critical dip to make less steep and ensure no diversion potential exists upon upgrading the crossing – apply 1.5" base rock through dip.
26.1	1. Rip and reshape 300m road segment to promote better surface drainage via crowning and camber. 2. 2. Apply 100 tons of 1.5"- rock base to armor road running surface
29	1. Replace existing pipe with new 42" x 60' L CMP with outlet set at the base of the OBF. Road width may be reduced if needed for installation. 2. Install critical dip near left hinge of Site #28 to draw flow towards inlet side. 3. Armor headcuts above inlet with 10 tons of 1-2 ft diameter rock. 4. Armor headcuts below outlet with 10 tons of 1-2 ft diameter rock. 5. 5. Install 1 rolling dip up left road.
33	Replace existing culvert with 36" x 30' CMP
34	1. Replace existing CMP with 60" x 40' CMP set with outlet placed at base of the OBF. 2. Armor OBF and inboard fill (IBF) with 18 tons of 1' – 2' dia. rock. 3. 3. Ensure no diversion potential exists upon upgrade.
--	**Quarry site.** This is not a treatment site. Previously stockpiled rock here may be used for project. There is approximately 40 tons of material available. Only existing stockpiled material may be used. No new excavation of the quarry is permitted. Coordinates: Latitude: 39.951746° Longitude: -123.648710°
70	1. Install 1 rolling dip ~ 60' left of DRC at gully 2. Install 40 tons of 0.5' - 2' diameter rock armor to the OBF below the rolling dip. 3. Install 1 rolling dip up the right road ~ 60' past driveway. 4. Armor rolling dip with 20 tons of 0.5' – 2' diameter rock on OBF 5. 5. Install 24"x 60' CMP at base of OBF
80	1. Install 1.5" base rock 85' through site at depth of 4" (16 tons) 2. 2. Replace existing CMP with 48" x 50' CMP (ensure >3" grade)

Site #	Treatment
81 (spur 26)	1. Install two rolling dips on each side of valley segment
99 (spur 26)	1. Install 54” x 30’ CMP at site, re-align road, shape failed outslope fill prism 2. Place 33 tons 1.5” base rock through realigned site for 100’
95 (Spur 26)	1. Decommission upper crossing site to restore streamflow to natural channel. Use spoils to fill diversion gully. Trackwalk to compact in 12” lifts. Provide for UTV access through site by laying back sidewalls to a minimum of 0.5x1.0. 2. Restore drainage through site, install 54”x40’ CMP, rock armor outfall with 8 tons 0.5’ – 2’ diameter rock, reshape outboard fill along 100’. 3. Install critical dip 4. Restore ditch and culvert to left of crossing by excavating inlet.
ADDITIVE LINE ITEMS	
30	1. Construct 3 rolling dips armored with 3” ballast rock and capped with the 1.5” base rock. This site requires an additional 15 tons of 3” ballast rock to secure subgrade in unstable terrain. 2. Install additional 60 tons of 1.5” base rock to secure site through 300 yards
77	1. Construct/reconstruct 2 rolling dips through site. 2. Install additional 60 tons of 1.5” base rock to secure site through 300 yards 3. Shape road prism to drain.

Table 2. Materials for site treatments.

Material Quantities Total (All Tasks)		Volume (CY)	Volume (tons)
1.5" Base Rock		348	512
6”-24" Armor Rock		90	140
3" Ballast Rock for dips		18	27
24” CMP 12 gauge aluminized steel	130 feet		
36” CMP 12 gauge aluminized steel	30 feet		
42” CMP 12 gauge aluminized steel	60 feet		
48” CMP 10 gauge aluminized steel	90 feet		
54” CMP 10 gauge aluminized steel	130 feet		
Treatment Site Location	**Dimensions**	**Volume (CY)**	**Volume (tons)**
Treatment Site 2			
Replace culvert with 54” x 60’		--	--
Rock for critical dip (1.5” base rock)	120' x 12' x 0.3’	21	32
Dip ballast rock (3”)	8 x 12’ x 0.5’	2	3
Treatment Site 24			
Replace culvert with 48” x 40’		--	--
OBF armor 0.5’ -1.5’ rock	20’ x 15’ x 1.5’	20	30
. Install 3 rolling dips up the right road (1.5” base rock)	120’ x 12’ x 0.4’ (x3)	21 (x3)	90
Dip ballast rock (3”)	8 x 12’ x 0.5’ (x3)	2 (x3)	9
Treatment Site 25			
Replace culvert with 24” x 50’		--	--
… Reconstruct/rock critical dip to broaden . (1.5” base rock)	60’ x 12’ x 0.5’	13	20
Treatment Site 26.1			
Apply 1.5” base rock over 300’	300 x 12 x 0.5	67	100
Treatment Site 29			
Replace culvert with 42” x 60’		--	--
Install Critical dip (1.5” base rock)	120' x 12' x 0.3’	21	32
. Armor headcuts above and below with 0.5’ -1.5’ rock	20’ x 6’ x 1.5’ (x2)	6 (x2)	20
1 rolling dip install (1.5” base rock)	120’ x 12’ x 0.4’	21	32
Dip ballast rock (3”)	8 x 12’ x 0.5’	2	3
Need to id spoils site for obf removal	COR	--	--
Treatment Site 33			
Replace culvert with 36” x 30’			
Treatment Site 34			
Replace culvert with 60” x 40’		--	--

Armor OBF with 0.5’ -1.5’ rock	15’ x 10’ x 1.5’	8	12
Armor IBF with 0.5’ -1.5’ rock	12’ x 6’ x 1.5’	4	6
Treatment Site 70			
Replace culvert with 24” x 80’			
Install 2 rolling dips (1.5” base rock)	120’ x 12’ x 0.4’ (x2)	21 (x2)	60
Dip ballast rock (3”)	8 x 12’ x 0.5’ (x2)	2 (x2)	6
Obf armor below dips 0.5’ -1.5’ rock	12’ x 30’ x 1.5’ (x2)	20 (x2)	60
Treatment Site 80 (spur 26)			
Replace culvert with 48” x 50’		--	--
Install 85’ of 1.5” Base rock	85’ x 12’ x 0.4’	15	22
Treatment Site 81 (spur 26)			
Install 2 rolling dips (1.5” base rock)	120’ x 12’ x 0.4’ (x2)	21 (x2)	60
Dip ballast rock (3”)	8 x 12’ x 0.5’ (x2)	2 (x2)	6
Treatment Site 99 (spur 26)			
Replace culvert with 54” x 30’		--	--
Install 1.5” rock through site	100’ x 12’ x 0.5’	22	33
Treatment Site 95 (spur 2)			
Replace culvert with 54” x 40’			
Rock armor OBF	12’ x 8’ x 1.5 ‘	5	8
Install critical dip (1.5” base rock)	120' x 12' x 0.3’	21	32
ADDITIVE LINE ITEMS			
Additive Site 30			
Install 3 rolling dips through site	120’ x 12’ x 0.4’ (x3)	21(3)	90
Additional 3” ballast rock			15
. Install additional 60 tons of base rock through site			60
Additive Site 77			
Install 2 rolling dips through site	120’ x 12’ x 0.4’ (x2)	21 (x2)	60
. Install additional 60 tons of base rock . through site			60

Environmental Controls. Federal and state permitting requirements have mandated project start and end dates from May 15 through October 15. Operations may extend past October 15th only if weather and moisture conditions are suitable and subject to CO approval. The Best Management Practices (BMPs) listed in Appendix B are required to satisfy Clean Water Act requirements for Non-point source pollution. The presence of archaeological resources, sensitive soils and private lands within the project area requires that all vehicles and equipment operate on the existing roadway and up to 50 feet from the road centerline as required to complete work at designated sites. Only previously stock-piled and disturbed rock may be used with the quarry site (see Government Furnished Property Section below)

Site Use Plan. All vehicle and equipment use shall be restricted to the existing roadway. Equipment work may occur within 50 feet of the road centerline if required for culvert replacements and rock installation. Equipment operations on areas beyond 50 feet require approval from the Contracting Officer (CO) or the COR. Existing pull-outs may be used for equipment staging and materials storage. Other sites may be utilized for staging, storage or disposal of excess soils subject to the Contracting Officer or the COR's approval. The quarry site at milepost 11.4 may be utilized for materials storage and staging. At the conclusion of work, all storage and staging areas shall be restored to a free-draining condition and be free of any inorganic debris. All excess soils will be groomed to match existing topography as feasible. No hazardous materials or fuels shall be stored within 150 feet of a watercourse.

Traffic control. Contractor shall be responsible for maintaining traffic control at all treatment sites. Red Mountain Road is a low use rural road and does not require frequent traffic control. However, infrequent residential and recreational access use does occur on the road. Roads should remain accessible during non-work periods using metal plates or other means when culvert installations are in progress. The contractor shall be responsible for notifying relevant road users of access status. Contacts will be provided by the COR.

Warranties. The contractor shall provide a 1-year warranty that work performed under this contract conforms to the contract requirements and is free of any defect in equipment, material, or design furnished, or workmanship performed by the Contractor or any subcontractor or supplier at any tier. The warranty provisions shall adhere to FAR 52.246-21.

5.0 Delivery

The Contractor shall complete all treatments within 100 calendar days after notice to proceed. Due to permitting requirements for the protection of water quality and federally listed species, all work must occur between May 15 and October 15. Work may occur after October 15 only if suitable weather and moisture conditions are forecast and with approval from the CO.

Final Inspection and Acceptance. The COR will review all work. At the conclusion of the work, the CO, or their representative, will inspect all sites for conformance with the specifications and requirements. Any deficiencies noted during the inspection will be corrected prior to release of final payment. Prior to final payment the Contractor will have properly disposed of all waste materials including culverts, completed site mulching and grooming of staging and storage areas.

6.0 Government-Furnished Property

The government may furnish up to 30 tons of unprocessed quarry rock from a quarry site at approximately milepost 11.4 along Red Mountain Road. The site is displayed in the attached map and listed in the treatment table.

The BLM will stake and flag the site prior to work. The Contractor shall be responsible for processing and hauling this material to the treatment site(s). Materials may only be used for treatment sites along Red Mountain Road. Unused materials shall be left onsite and the site groomed to restore a more natural appearance. Any excess rock from the project may be staged and stored at the quarry site. Only previously worked and disturbed materials may be utilized.

7.0 Place of Performance

The work shall occur on Red Mountain Road in Mendocino County, California near the community of Piercy. See Maps 1 and 2. Work will commence at approximately milepost 2.5 and extend to mile 12. The BLM will mark these start and endpoints prior to work.

Coordinates for these points are:

- Intersection of Red Mountain Road and State Highway 271
    - Latitude: 39.946792°
    - Longitude: -123.779338°
- Start of 12 mile treatment segment at mile 1.9
    - Latitude: 39.951553°
    - Longitude: -123.757409°
- End of 12 mile treatment segment at mile 12
    - Latitude: 39.919714°
    - Longitude: -123.644183°

Red Mountain Road is located in northernmost Mendocino County near the community of Piercy. The road is accessed at exit 625 off of Highway 101.

Coming from the north on Highway 101, exit 625 is approximately 13.5 miles south of Garberville and 1.5 south of the Piercy exit. At the exit, turn left and proceed 0.2 miles to the Red Mountain Road intersection which is signed.

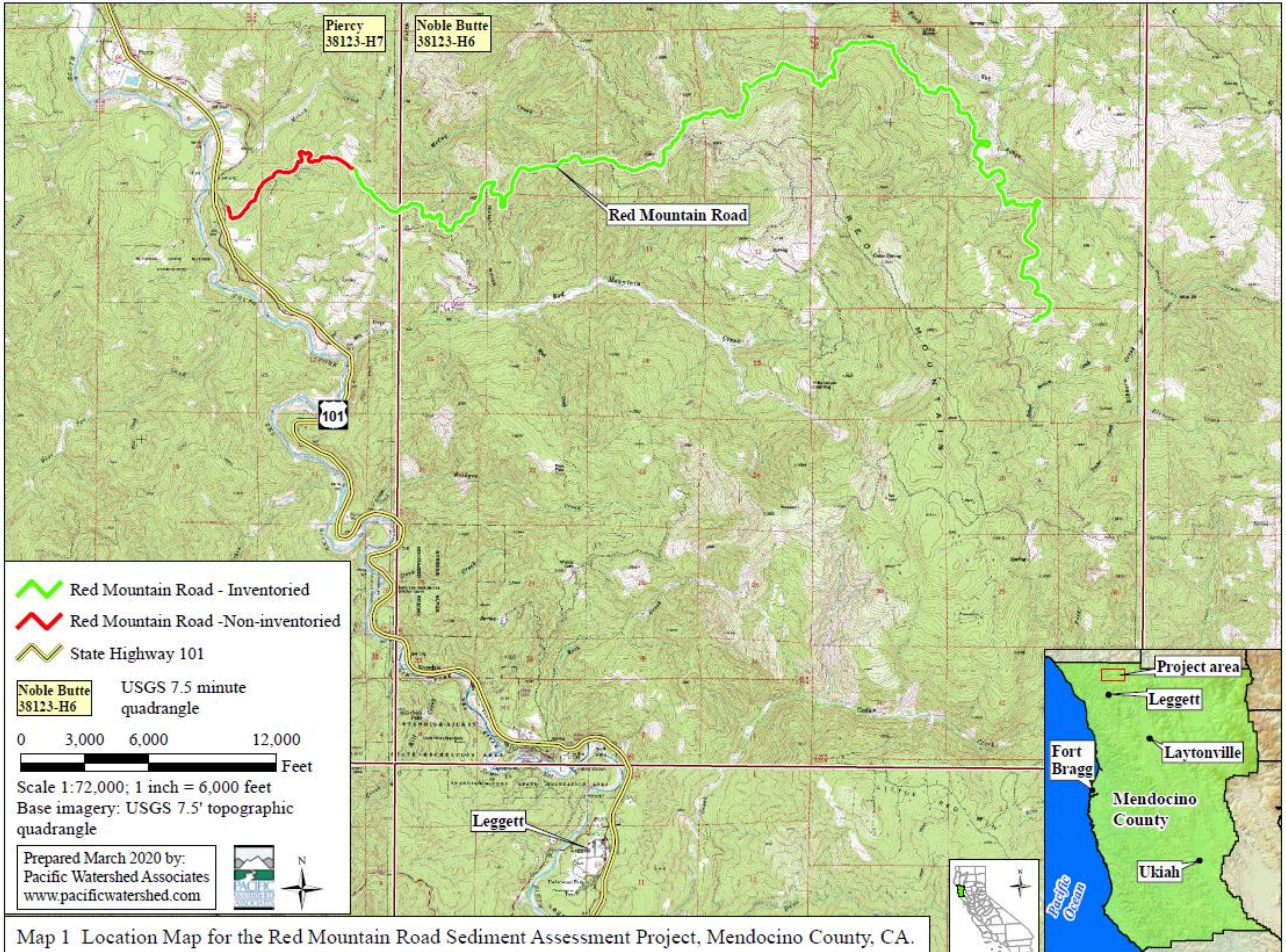
Coming from the south on Highway 101, exit 625 is approximately 13 miles north of Leggett and 2 miles north of the Confusion Hill roadside park. At the exit, turn right and proceed 0.1 miles to the Red Mountain Road intersection.

Red Mountain Road traverses approximately 12 miles of private and public lands. <u>Moderate to high clearance vehicles are recommended for traveling the road.</u> Any milepost figures for treatment sites are referenced from the intersection of Red Mountain Road and Highway 271 (“from the pavement”).

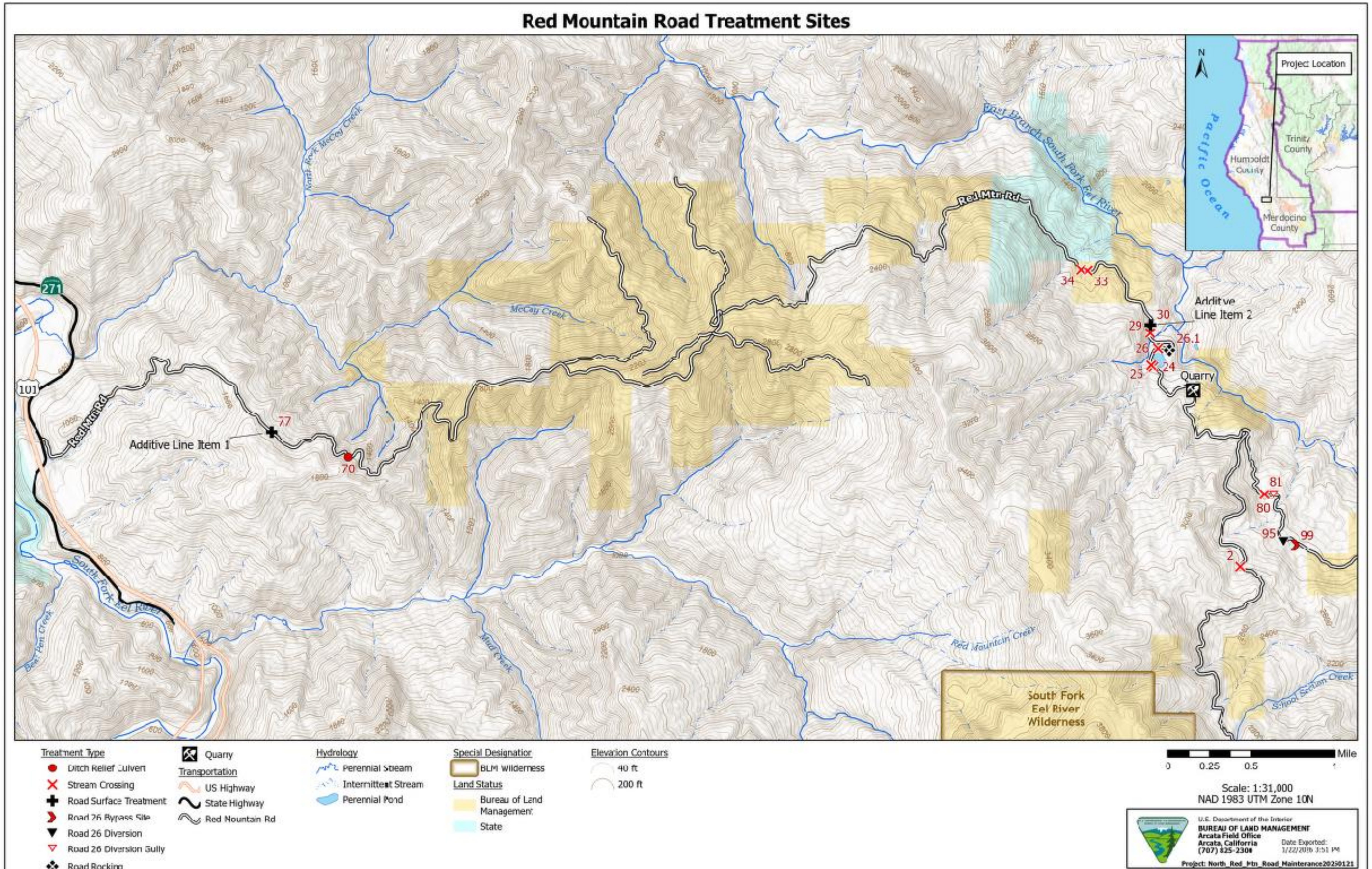
8.0 Period of Performance

The period of performance will extend 100 calendar days from the Notice to Proceed date. Due to permitting requirements for the protection of water quality and federally listed species, all work must occur between May 15 and October 15. An extension may be granted for work past October 15 after review of weather and moisture conditions and approval from the CO.

Map 1. Red Mountain Road General Location



Map 2. Red Mountain Road Treatment Sites.



Appendix A

Schematics for Typical Treatments
Specified in the Scope of Work
and detailed in the Site-Specific Treatments Table.

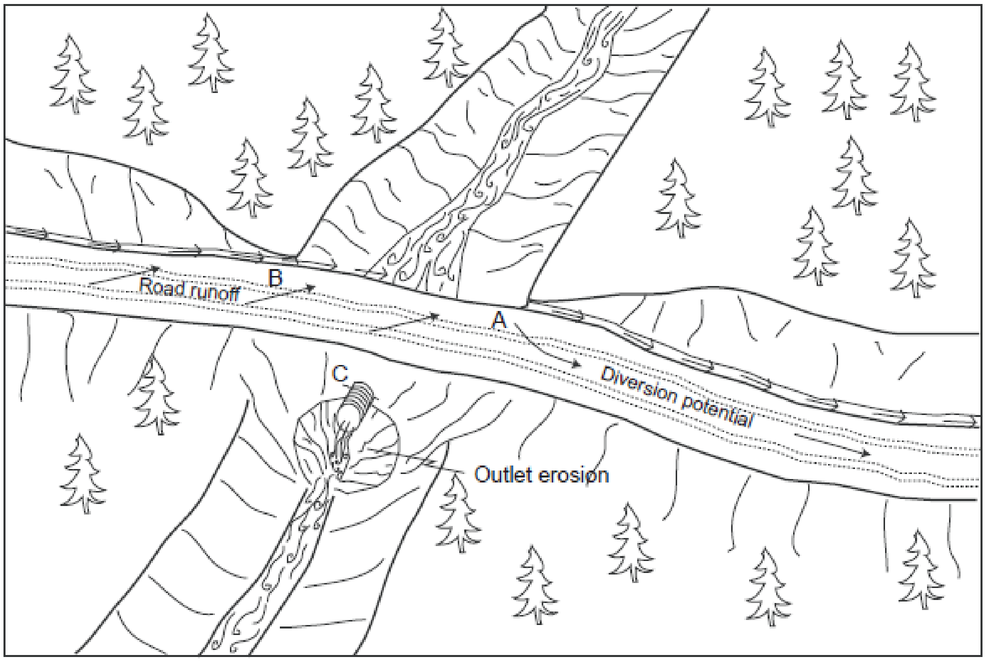
Typical Problems and Applied Treatments for a Non-fish Bearing Upgraded Stream Crossing

Problem condition (before)

A - Diversion potential

B - Road surface and ditch drain to stream

C - Undersized culvert high in fill with outlet erosion

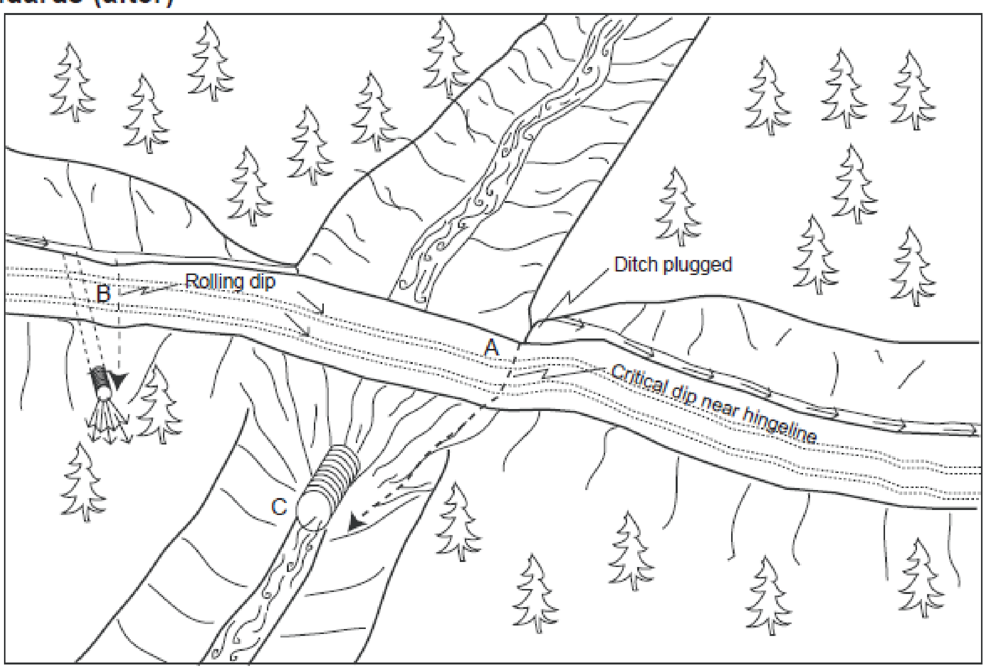


Treatment standards (after)

A - No diversion potential with critical dip installed near hingeline

B - Road surface and ditch disconnected from stream by rolling dip and ditch relief culvert

C - 100-year culvert set at base of fill



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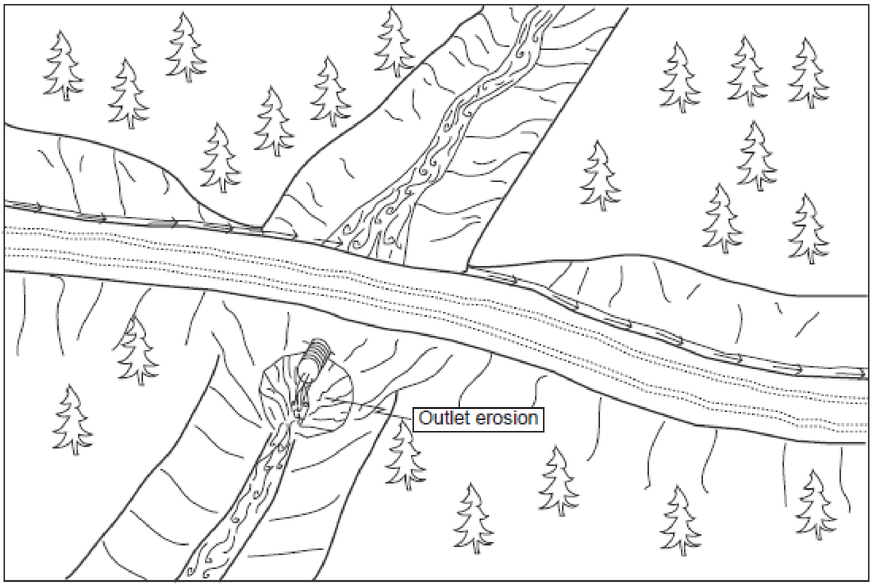
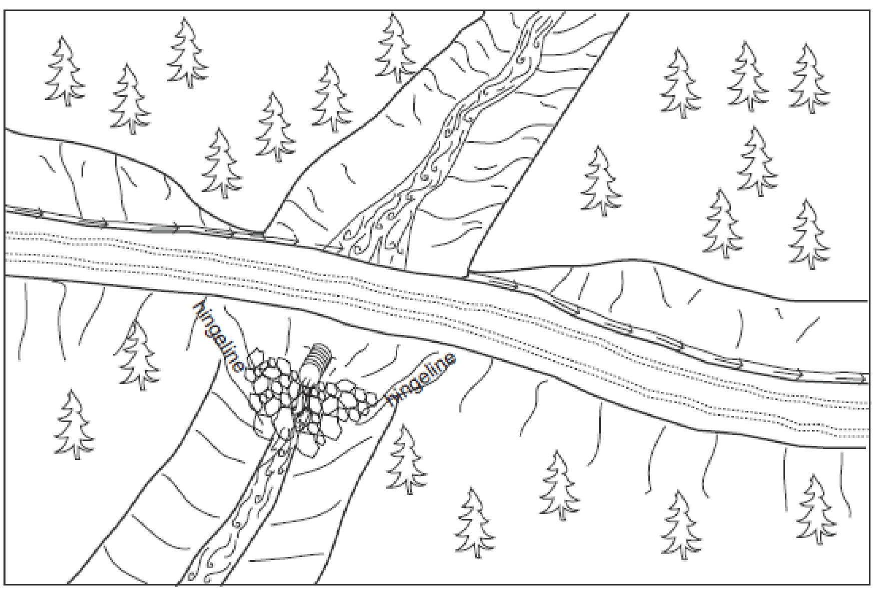
PWA Typical Drawing #1a

Armoring Fill Faces to Upgrade Stream Crossings



Problem: Culvert set high in outboard fill has resulted in scour of the outboard fill face and natural channel.
Conditions: The existing stream crossing has a culvert sufficient in diameter to manage design stream flows and has a functional life.



Action: The area of scour is backfilled with rip-rap to provide protection in the form of energy dissipation for the remaining fill face and channel.

Treatment Specifications:

1) Placement of rip-rap should be between the left and right hingelines and extend from a keyway excavated below the existing channel base level at the base of the fill slope up and under the existing culvert.
2) Rock size and volume is determined on a site by site basis based on estimated discharge and existing stream bed particle size range (See accompanying road log).

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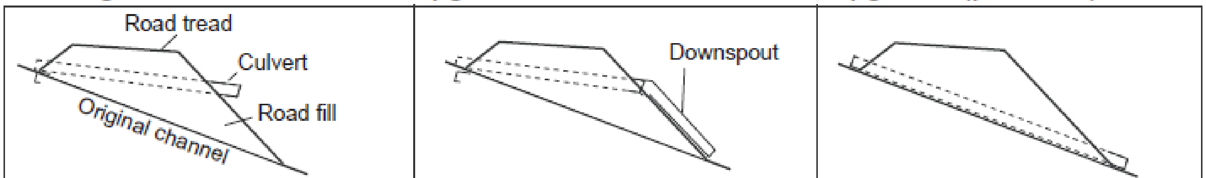
PWA Typical Drawing #1b

Typical Design of a Non-fish Bearing Culverted Stream Crossing



Existing

1. Culvert not placed at channel grade.
2. culvert does not extend past base of fill.

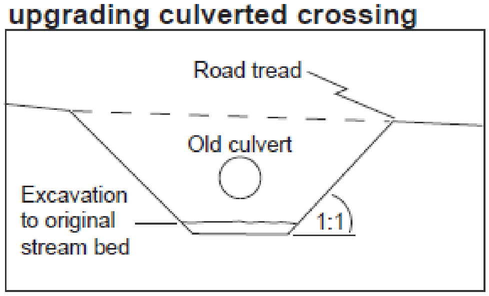
Upgraded

1. Culvert not placed at channel grade.
2. Downspout added to extend outlet past road fill.

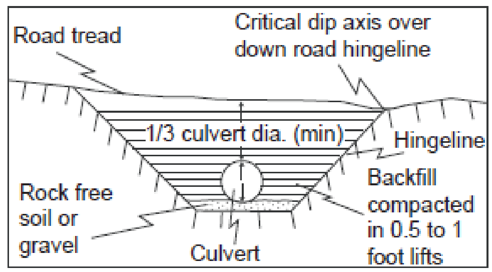
Upgraded (preferred)

1. Culvert placed at channel grade.
2. Culvert inlet and outlet rest on, or partially in, the originial streambed.

Excavation in preparation for upgrading culverted crossing



Upgraded stream crossing culvert installation



Note:
Road upgrading tasks typically include upgrading stream crossings by installing larger culverts and inlet protection (trash barriers) to prevent plugging. Culvert sizing for the 100-year peak storm flow should be determined by both field observation and calulations using a procedure such as the Rational Formula.

Stream crossing culvert Installation

1. Culverts shall be aligned with natural stream channels to ensure proper function, and prevent bank erosion and plugging by debris.
2. Culverts shall be placed at the base of the fill and the grade of the original streambed, or downspouted past the base of the fill.
3. Culverts shall be set slightly below the original stream grade so that the water drops several inches as it enters the pipe.
5. To allow for sagging after burial, a camber shall be between 1.5 to 3 incher per 10 feet culvert pipe length.
6. Backfill material shall be free of rocks, limbs or other debris that could dent or puncture the pipe or allow water to seep around pipe.
7. First one end then the other end of the culvert shall be covered and secured. The center is covered last.
8. Backfill material shall be tamped and compacted throughout the entire process:
   - Base and side wall material will be compacted before the pipe is placed in its bed.
   - Backfill compacting will be done in 0.5 - 1 foot lifts until 1/3 of the diameter of the culvert has been covered. A gas powered tamper can be used for this work.
9. Inlets and outlets shall be armored with rock or mulched and seeded with grass as needed.
10. Trash protectors shall be installed just upstream from the culvert where there is a hazard of floating debris plugging the culvert.
11. Layers of fill will be pushed over the crossing until the final designed road grade is achieved, at a minimum of 1/3 to 1/2 the culvert diameter.

Erosion control measures for culvert replacement

Both mechanical and vegetative measures will be employed to minimize accelerated erosion from stream crossing and ditch relief culvert upgrading. Erosion control measures implemented will be evaluated on a site by site basis. Erosion control measures include but are not limited to:

1. Minimizing soil exposure by limiting excavation areas and heavy equipment distrubance.
2. Installing filter windrows of slash at the base of the road fill to minimize the movement of eroded soil to downslope areas and stream channels.
3. Retaining rooted trees and shrubs at the base of the fill as "anchor" for the fill and filter windrows.
4. Bare slopes created by construction operations will be protected until vegetation can stabilize the surface. Surface erosion on exposed cuts and fills will be minimized by mulching, seeding, planting, compacting, armoring, and/or benching prior to the first rains.
5. Excess or unusable soil will be stored in long term spoil disposal locations that are not limited by factors such as excessive moisture, steep slopes greater than 10%, archeology potential, or proximity to a watercourse.
6. On running streams, water will be pumped or diverted past the crossing and into the downstream channel during the construction process.
7. Straw bales and/or silt fencing will be employed where necessary to control runoff within the construction zone.

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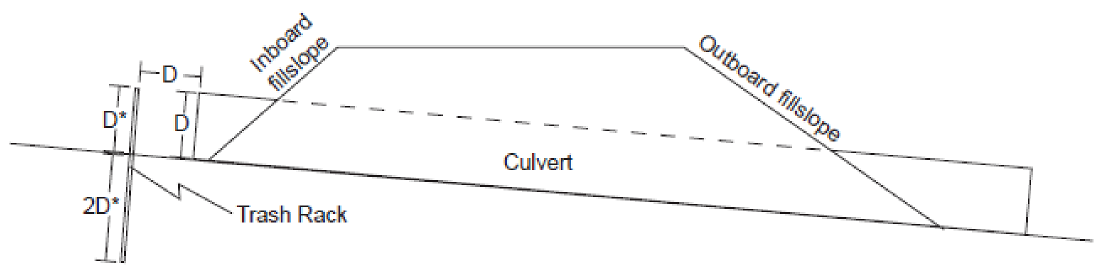
Typical Drawing #2

Typical Design of a Single-post Culvert Inlet Trash Rack

Cross section view



D - Culvert diameter

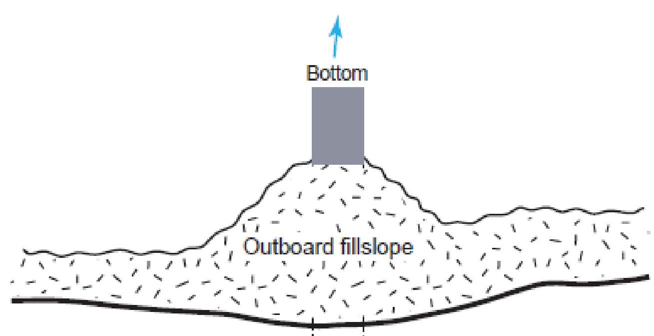
D* - If the culvert is designed for the 100-year peak storm flow, the trash rack height above the streambed should equal D.
If the culvert is undersized, then the trash rack needs to be extended vertically above the streambed to match or exceed the expected headwall height.

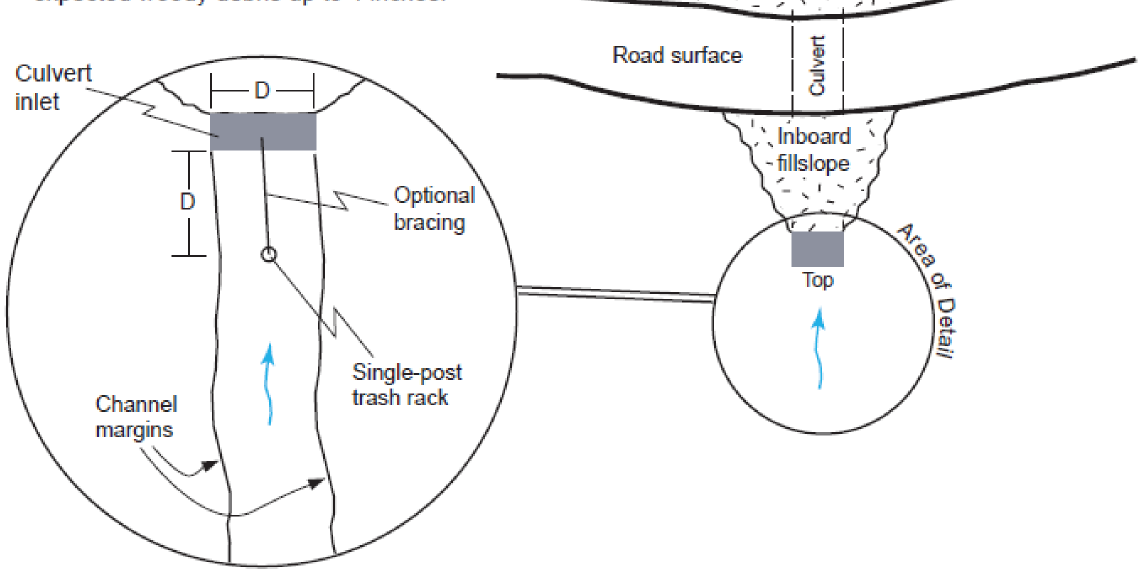
Plan view

Notes:

1. Many materials can be used for a single-post trash rack including old railroad track, galvanized pipe, and fence posts.

2. The diameter of single-post trash racks should be sized based on the size of expected woody debris. As a basic rule of thumb, the diameter of the trash rack should be equal to the diameter of the expected woody debris up to 4 inches.





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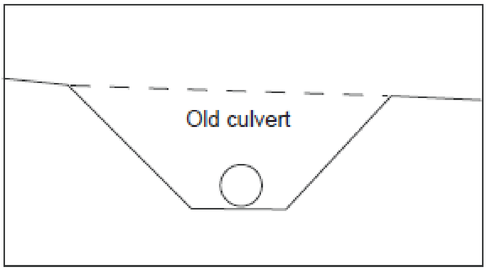
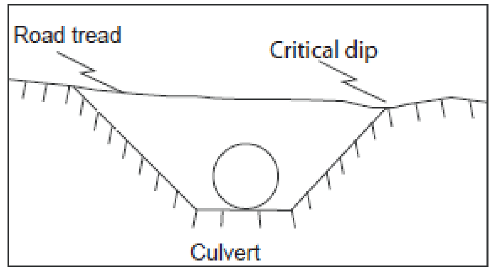
Typical Drawing #3

Typical Design of Upgraded Stream Crossings





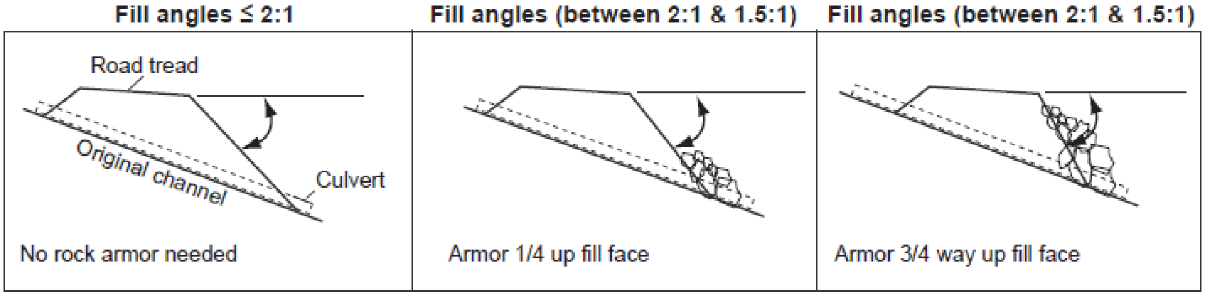
Stream crossing culvert Installation

1. Culverts shall be aligned with natural stream channels to ensure proper function, and prevent bank erosion and plugging by debris.
2. Culverts shall be placed at the base of the fill and the grade of the original streambed or downspouted past the base of the fill.
3. Culverts shall be set slightly below the original stream grade so that the water drops several inches as it enters the pipe.
5. To allow for sagging after burial, a camber shall be between 1.5 to 3 incher per 10 feet culvert pipe length.
6. Backfill material shall be free of rocks, limbs or other debris that could dent or puncture the pipe or allow water to seep around pipe.
7. First one end and then the other end of the culvert shall be covered and secured. The center is covered last.
8. Backfill material shall be tamped and compacted throughout the entire process:
   - Base and side wall material will be compacted before the pipe is placed in its bed.
   - backfill compacting will be done in 0.5 - 1 foot lifts until 1/3 of the diameter of the culvert has been covered. A gas powered tamper can be used for this work.
9. Inlets and outlets shall be armored with rock or mulched and seeded with grass as needed.
10. Trash protectors shall be installed just upstream from the culvert where there is a hazard of floating debris plugging the culvert.
11. Layers of fill will be pushed over the crossing until the final designed road grade is achieved, at a minimum of 1/3 to 1/2 the culvert diameter.

Note:
Road upgrading tasks typically include upgrading stream crossings by installing larger culverts and inlet protection (trash barriers) to prevent plugging. Culvert sizing for the 100-year peak storm flow should be determined by both field observation and calculations using a procedure such as the Rational Formula.

Armoring fill faces

Fill angles ≤ 2:1 | **Fill angles (between 2:1 & 1.5:1)** | **Fill angles (between 2:1 & 1.5:1)**



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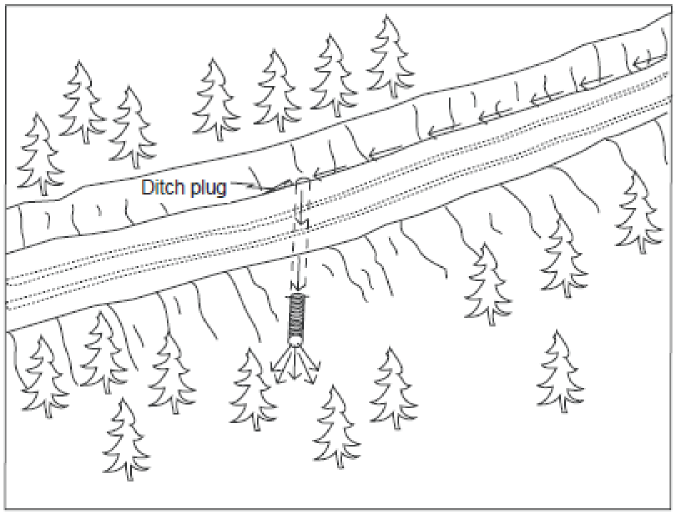
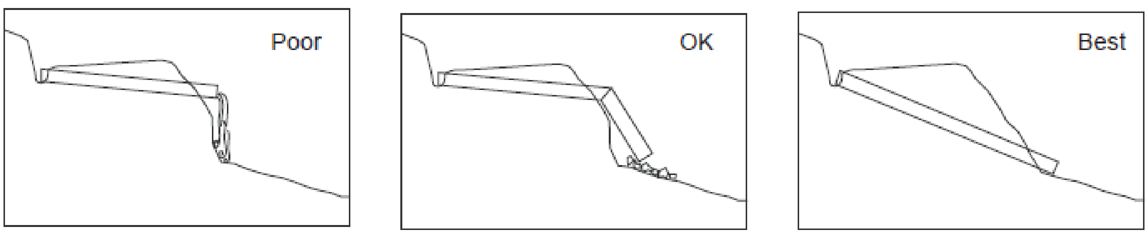
PWA Typical Drawing #4

Typical Ditch Relief Culvert Installation





Ditch relief culvert installation

1) The same basic steps followed for stream crossing installation shall be employed.
2) Culverts shall be installed at a 30 degree angle to the ditch to lessen the chance of inlet erosion and plugging.
3) Culverts shall be seated on the natural slope or at a minimum depth of 5 feet at the outside edge of the road, whichever is less.
4) At a minimum, culverts shall be installed at a slope of 2 to 4 percent steeper than the approaching ditch grade, or at least 5 inches every 10 feet.
5) Backfill shall be compacted from the bed to a depth of 1 foot or 1/3 of the culvert diameter, which ever is greater, over the top of the culvert.
6) Culvert outlets shall extend beyond the base of the road fill (or a flume downspout will be used). 7Culverts will be seated on the natural slope or at a depth of 5 feet at the outside edge of the road, whichever is less.

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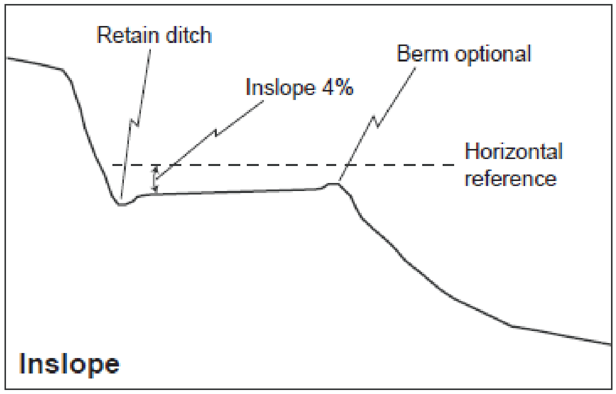
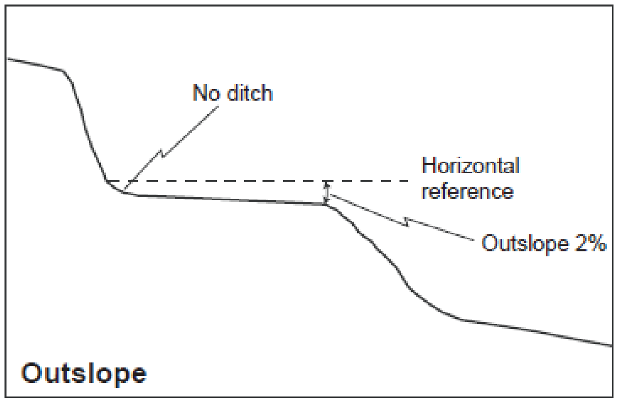
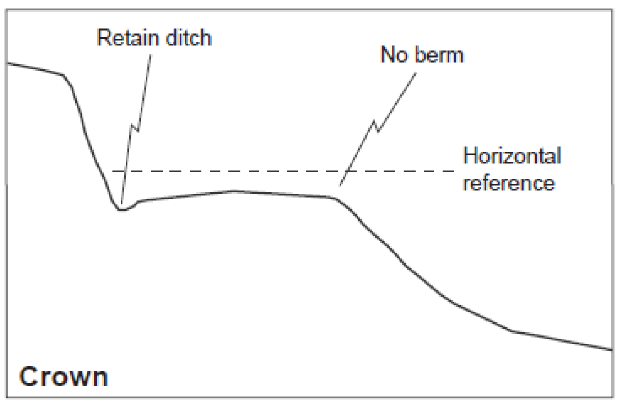
Typical Drawing #8

Typical Designs for Using Road Shape to Control Road Runoff







Outsloping Pitch for Roads Up to 8% Grade		
Road grade	**Unsurfaced roads**	**Surfaced roads**
4% or less	3/8" per foot	1/2" per foot
5%	1/2" per foot	5/8" per foot
6%	5/8" per foot	3/4" per foot
7%	3/4" per foot	7/8" per foot
8% or more	1" per foot	1 1/4" per foot

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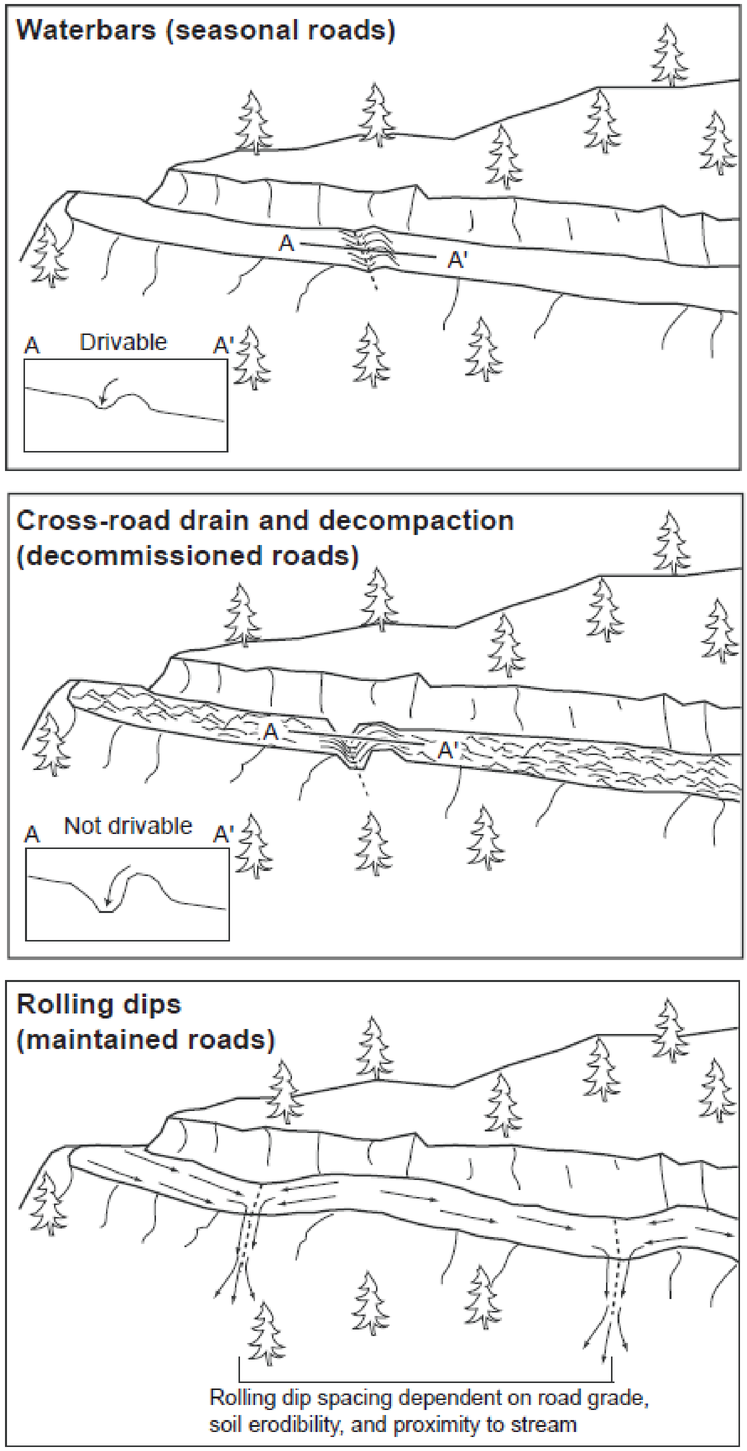
Typical Drawing #9

Typical Methods for Dispersing Road Surface Runoff with Waterbars, Cross-road Drains, and Rolling Dips



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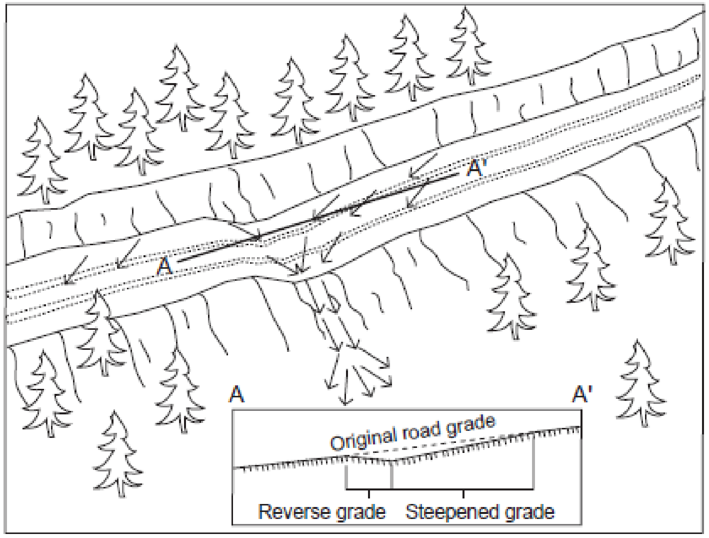
Typical Drawing #10

Typical Road Surface Drainage by Rolling Dips



Rolling dip installation:

1. Rolling dips will be installed in the roadbed as needed to drain the road surface.
2. Rolling dips will be sloped either into the ditch or to the outside of the road edge as required to properly drain the road.
3. Rolling dips are usually built at 30 to 45 degree angles to the road alignment with cross road grade of at least 1% greater than the grade of the road.
4. Excavation for the dips will be done with a medium-size bulldozer or similar equipment.
5. Excavation of the dips will begin 50 to 100 feet up road from where the axis of the dip is planned as per guidelines established in the rolling dip dimensions table.
6. Material will be progressively excavated from the roadbed, steepening the grade unitl the axis is reached.
7. The depth of the dip will be determined by the grade of the road (see table below).
8. On the down road side of the rolling dip axis, a grade change will be installed to prevent the runoff from continuing down the road (see figure above).
9. The rise in the reverse grade will be carried for about 10 to 20 feet and then return to the original slope.
10. The transition from axis to bottom, through rising grade to falling grade, will be in a road distance of at least 15 to 30 feet.

Table of rolling dip dimensions by road grade

Road grade %	**Upslope approach distance** (from up road start to trough) ft	**Reverse grade distance** (from trough to crest) ft	**Depth at trough outlet** (below average road grade) ft	**Depth at trough inlet** (below average road grade) ft
<6	55	15 - 20	0.9	0.3
8	65	15 - 20	1.0	0.2
10	75	15 - 20	1.1	0.01
12	85	20 - 25	1.2	0.01
>12	100	20 - 25	1.3	0.01

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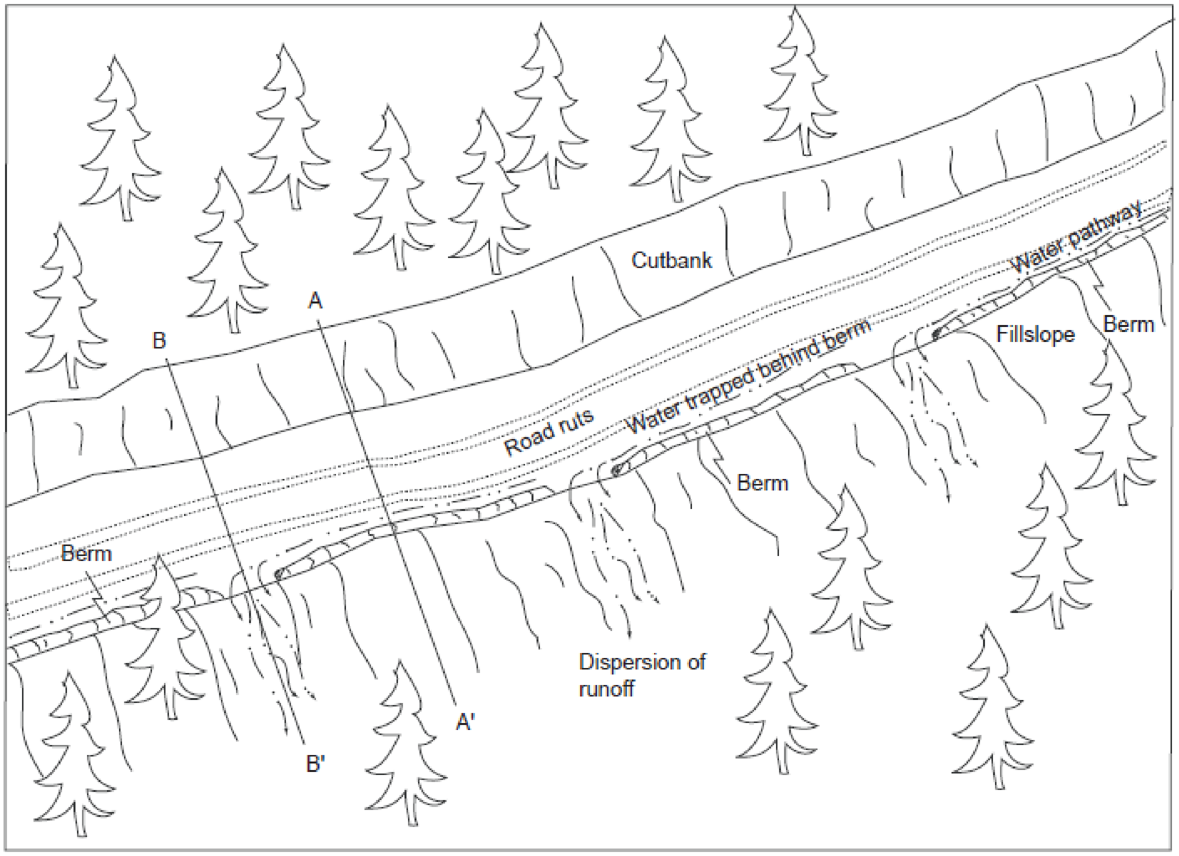
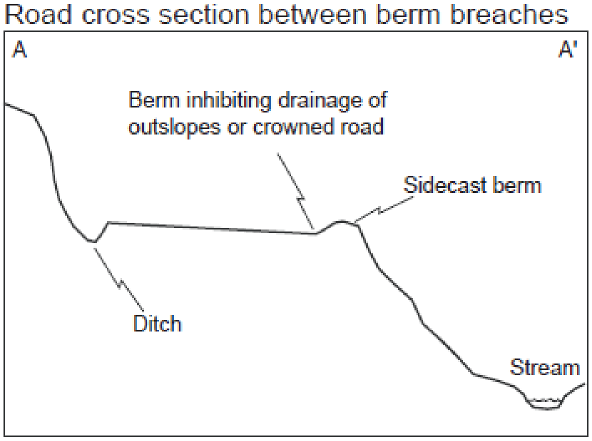
Typical Drawing #11

Typical Sidecast or Excavation Methods for Removing Outboard Berms on a Maintained Road

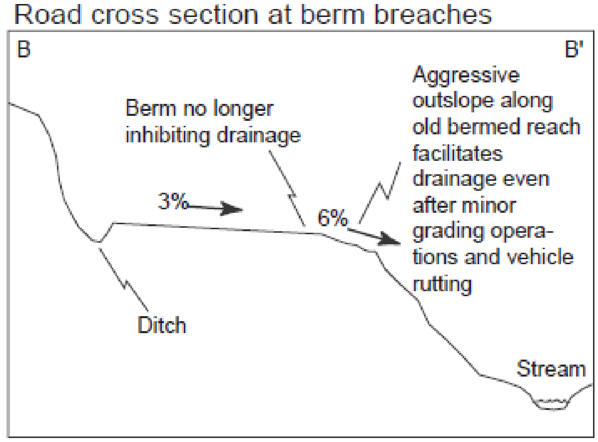
1. On gentle road segments berms can be removed continuously (see B-B').
2. On steep road segments, where safety is a concern, the berm can be frequently breached (see A-A' & B-B') Berm breaches should be spaced every 30 to 100 feet to provide adequate drainage of the road system while maintaining a semi-continuous berm for vehicle safety.



Road cross section between berm breaches



Road cross section at berm breaches



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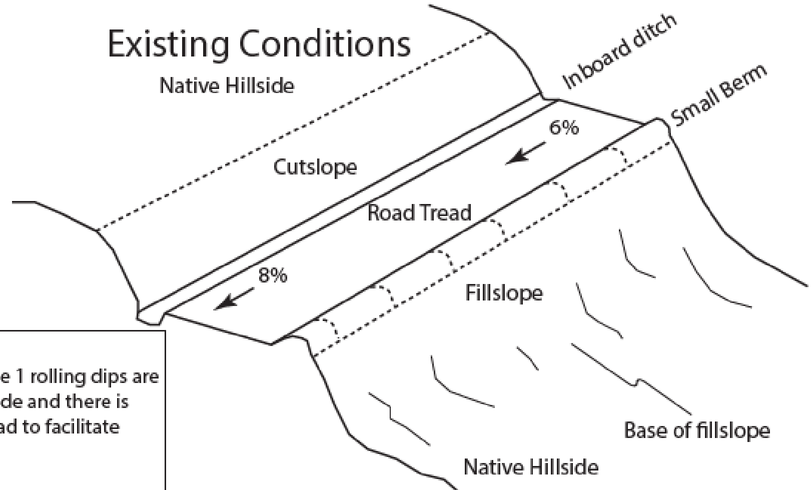
Typical Drawing #12

Standard (Type 1) Rolling Dip Construction

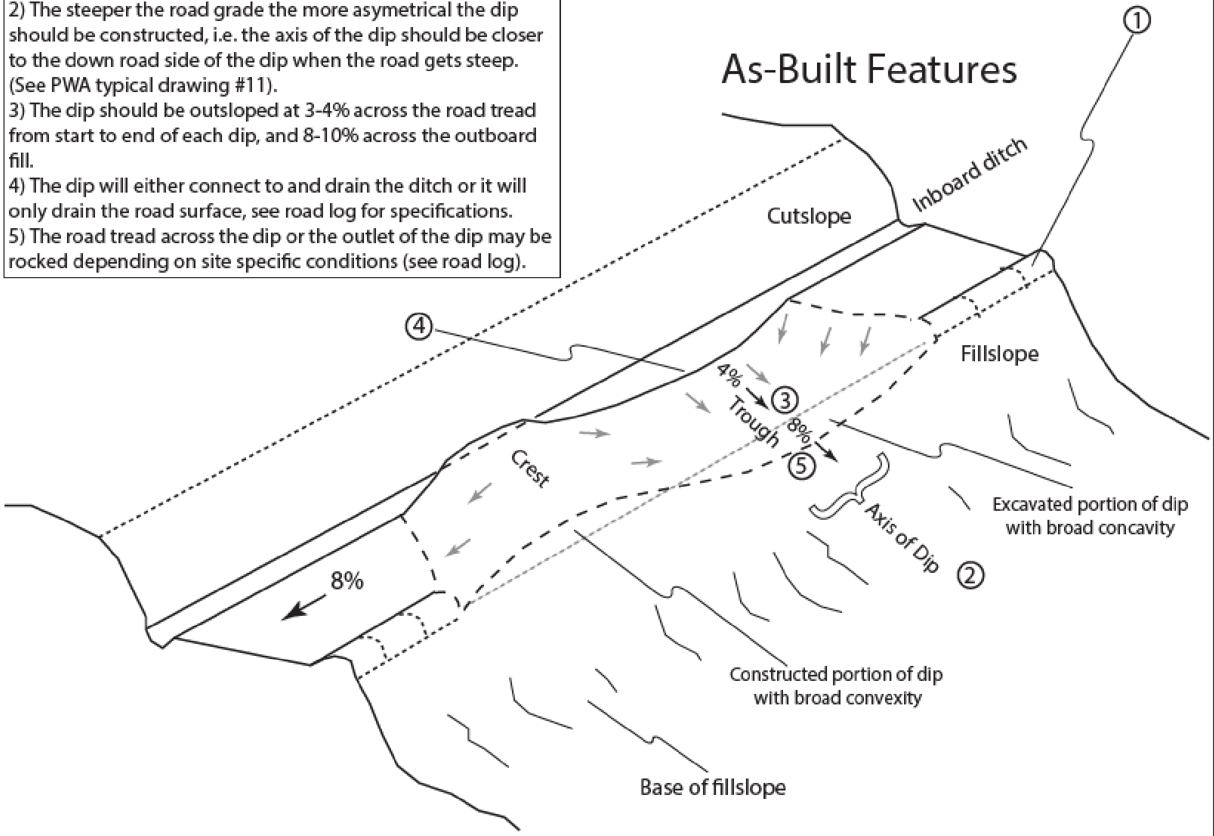


Notes

Rolling dip type 1 existing conditions: Type 1 rolling dips are utilized when roads are less than 12-14% grade and there is proximal outfall adjacent to the outboard road to facilitate road drainage.

Design Notes:

1) The berm should be removed for the entire length of the dip.
2) The steeper the road grade the more asymetrical the dip should be constructed, i.e. the axis of the dip should be closer to the down road side of the dip when the road gets steep. (See PWA typical drawing #11).
3) The dip should be outsloped at 3-4% across the road tread from start to end of each dip, and 8-10% across the outboard fill.
4) The dip will either connect to and drain the ditch or it will only drain the road surface, see road log for specifications.
5) The road tread across the dip or the outlet of the dip may be rocked depending on site specific conditions (see road log).



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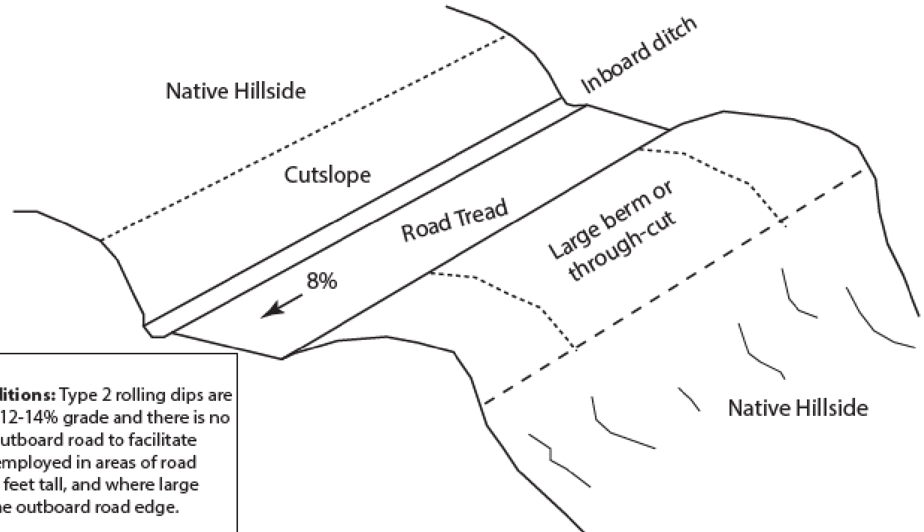
Typical Drawing #19a

Type 2 Rolling Dip Construction

(Through-cut or thick berm road reaches)



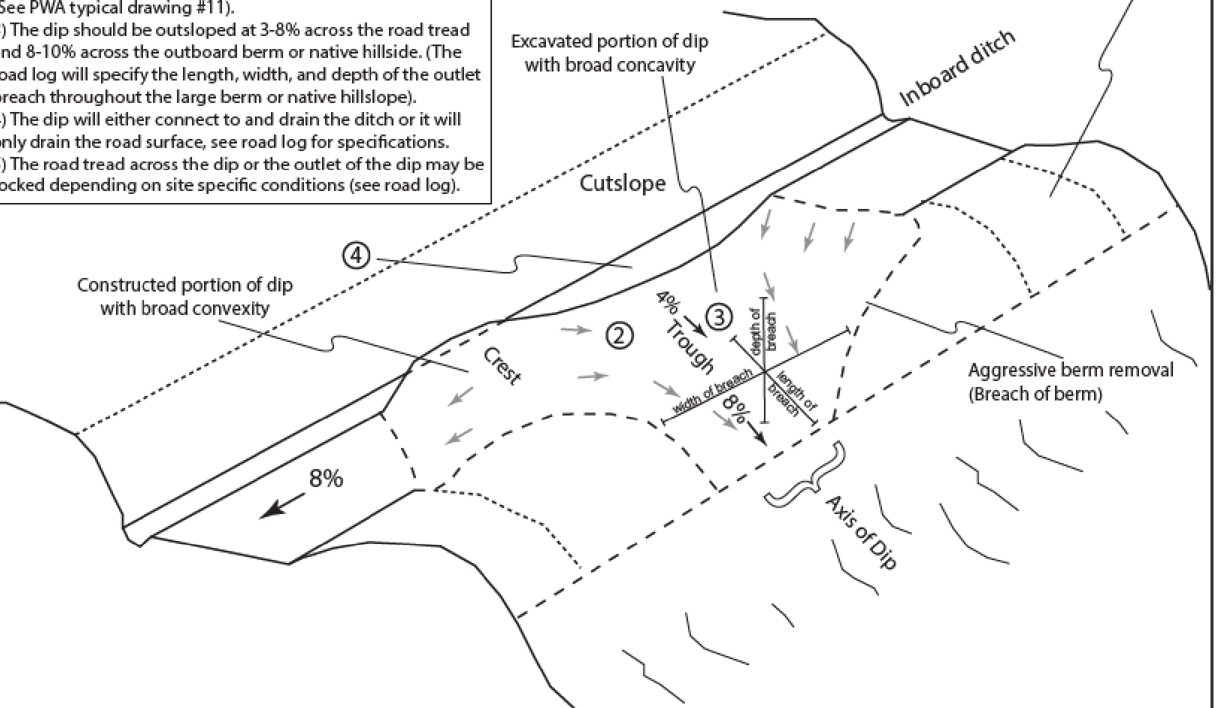
Notes

Rolling dip type 2 existing conditions: Type 2 rolling dips are utilized when roads are less than 12-14% grade and there is no proximal outfall adjacent to the outboard road to facilitate road drainage. These should be employed in areas of road throughcuts generally less than 3 feet tall, and where large wide and/or tall berms exist on the outboard road edge.

Design Notes:

1) The berm or native hillside should be removed for the entire length of the excavated portion of the dip, or, at a minimuim through the trough/axis of the dip.
2) The steeper the road grade the more asymetrical the dip should be constructed, i.e. the axis of the dip should be closer to the down road side of the dip when the road gets steep. (See PWA typical drawing #11).
3) The dip should be outsloped at 3-8% across the road tread and 8-10% across the outboard berm or native hillside. (The road log will specify the length, width, and depth of the outlet breach throughout the large berm or native hillslope).
4) The dip will either connect to and drain the ditch or it will only drain the road surface, see road log for specifications.
5) The road tread across the dip or the outlet of the dip may be rocked depending on site specific conditions (see road log).

As-built Features



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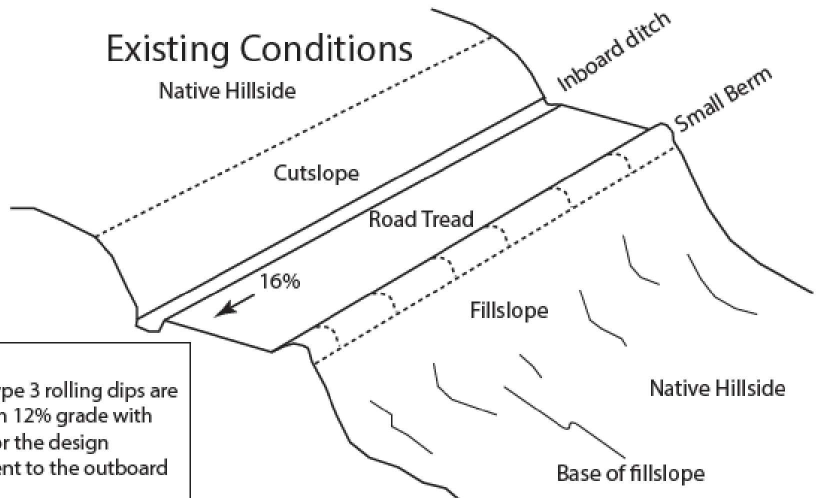
PWA Typical Drawing #19b

Type 3 Rolling Dip Construction
(steep slope outslope)

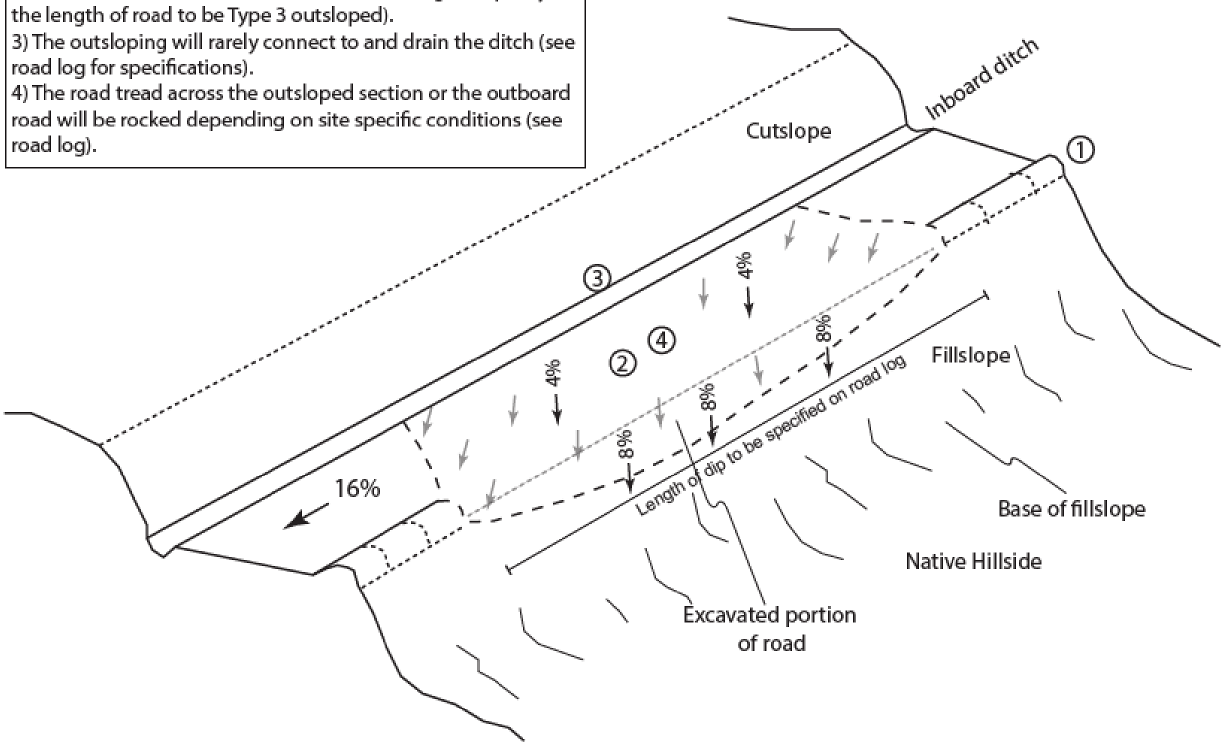


Notes

Rolling dip type 3 existing conditions: Type 3 rolling dips are utilized when roads grades are steeper than 12% grade with little opportunity to create reverse grade for the design vehicle, and there is proximal outfall adjacent to the outboard road to facilitate road drainage.

Design Notes:

1) The berm should be removed for the entire length of the outsloped section.
2) The dip should be outsloped at 3-8% across the road tread and 8-10% across the outboard fill. (The road log will specify the length of road to be Type 3 outsloped).
3) The outsloping will rarely connect to and drain the ditch (see road log for specifications).
4) The road tread across the outsloped section or the outboard road will be rocked depending on site specific conditions (see road log).



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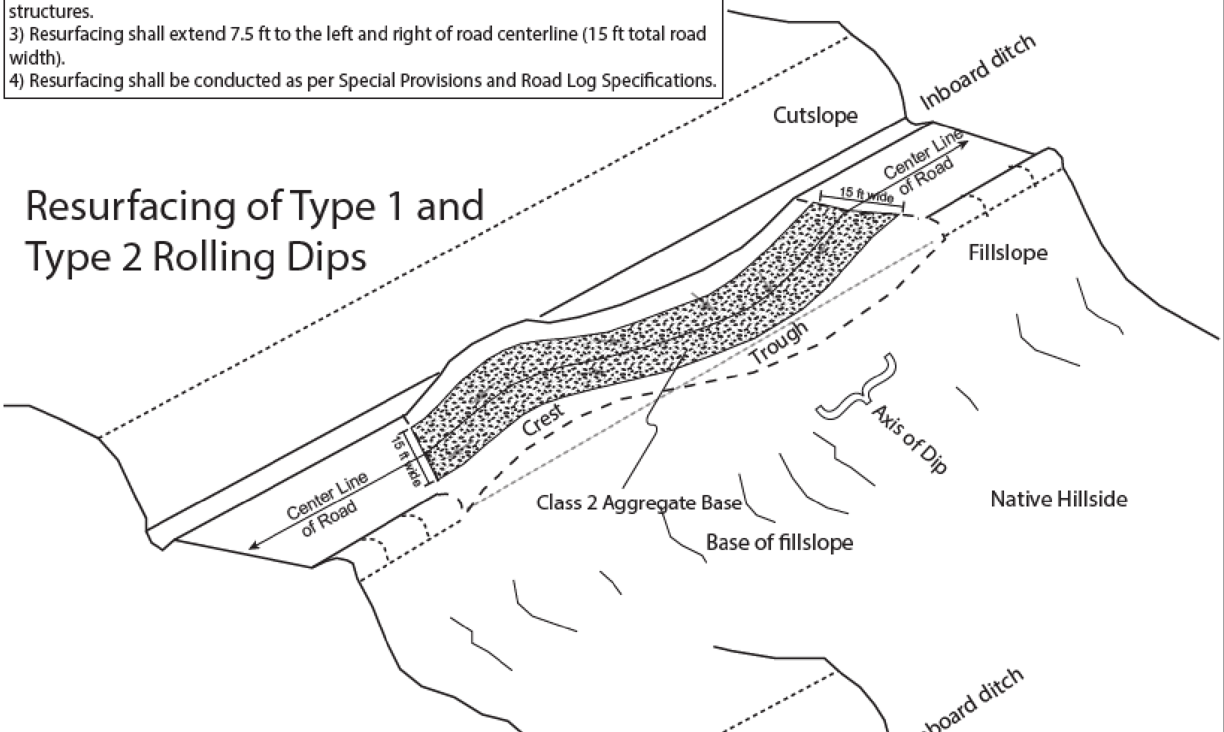
PWA Typical Drawing #19c

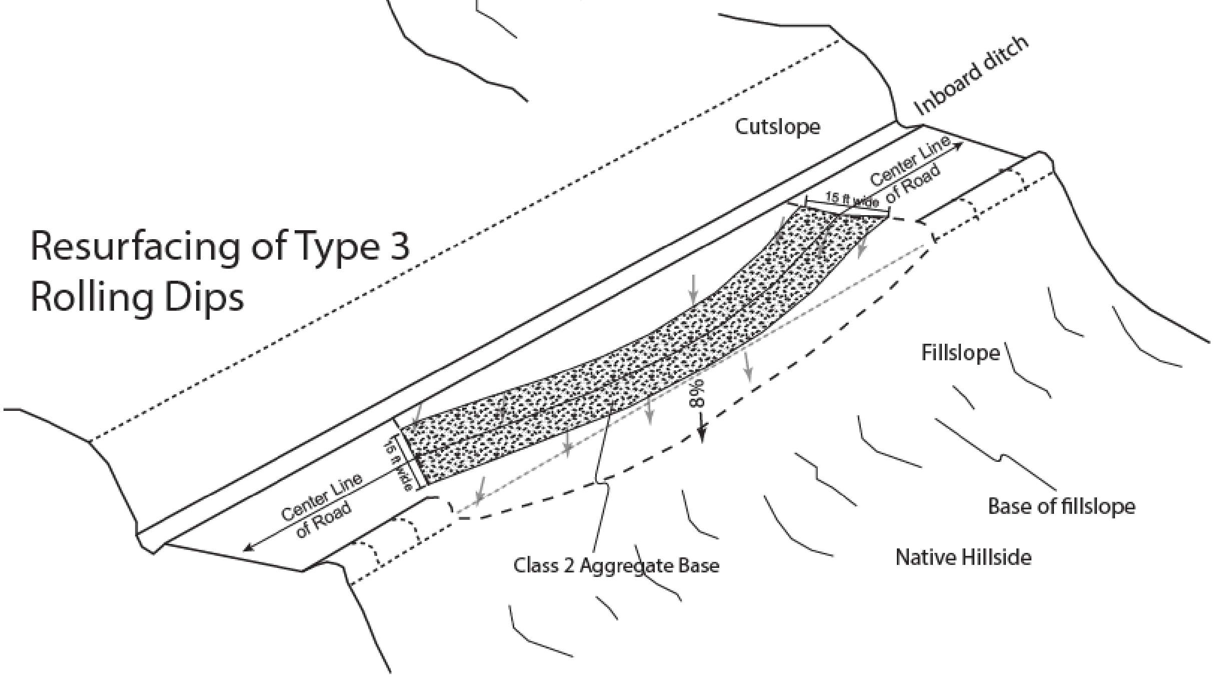
Resurfacing after Dip Construction

Notes
1) Road travelway areas disturbed by rolling dip construction shall be resurfaced with Class 2 Aggregate Base, as per Special Provisions.
2) Resurfacing shall extend through rolling dip excavated trough and constructed crest structures.
3) Resurfacing shall extend 7.5 ft to the left and right of road centerline (15 ft total road width).
4) Resurfacing shall be conducted as per Special Provisions and Road Log Specifications.

Resurfacing of Type 1 and Type 2 Rolling Dips



Resurfacing of Type 3 Rolling Dips



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Typical Drawing #22

Appendix B

Best Management Practices
For the Protection of Water Quality

To maintain compliance with federal Clean Water Act Requirements, Contractor is responsible for adhering to Best Management Practices. The following list of Best Management Practices (BMPs) will be employed at treatment sites depending on the nature of treatments and the applicability of a given BMP for the site.

Construction and reconstruction BMPs:

1. Minimize fill volumes at stream crossings by restricting width and height of fill to amounts needed for safe travel and adequate cover for culverts. For deep fills (generally greater than 15 feet deep), incorporate additional design criteria (e.g., rock blankets, buttressing, bioengineering techniques) to reduce the susceptibility of fill failures.
2. Place sediment-trapping materials or structures such as straw bales, jute netting, or sediment basins at the base of newly constructed fill or side slopes where sediment could be transported to waters of the State. Keep materials away from culvert inlets or outlets.
3. Construct road fills to prevent fill failure using inorganic material, compaction, buttressing, sub-surface drainage, rock facing, or other effective means.
4. Locate waste disposal areas outside wetlands, Riparian Reserve, floodplains, and unstable areas to minimize risk of sediment delivery to waters of the State. Apply surface erosion control prior to the wet season. Prevent overloading areas, which may become unstable.
5. Design stream crossings to minimize diversion potential in the event that the crossing is blocked by debris during storm events. This protection could include hardening crossings, armoring fills, dipping grades, oversizing culverts, hardening inlets and outlets, constructing critical dips, and lowering the fill height.
6. Design stream crossings to prevent diversion of water from streams into downgrade road ditches or down road surfaces.
7. On new construction, install culverts at the natural stream grade, unless a lessor gradient is required for fish passage.
8. Place instream grade control structures above or below the crossing structure, if necessary, to prevent stream head cutting, culvert undermining and downstream sedimentation. Employ bioengineering measures to protect the stability of the streambed and banks.
9. Prevent culvert plugging and failure in areas of active debris movement with measures such as beveled culvert inlets, flared inlets, wingwalls, over-sized culverts, trash racks, or slotted risers.

10. Utilize stream diversion and isolation techniques when installing stream crossings. Evaluate the physical characteristics of the site, volume of water flowing through the project area and the risk of erosion and sedimentation when selecting the proper techniques.
11. Disconnect road runoff to the stream channel by outsloping the road approach. If outsloping is not possible, use runoff control, erosion control and sediment containment measures. These may include using additional cross drain culverts, ditch lining, and catchment basins. Prevent or reduce ditch flow conveyance to the stream through cross drain placement above the stream crossing.

Surface Drainage BMPs:

1. Effectively drain the road surface by using crowning, insloping or outsloping, grade reversals (rolling dips), and waterbars or a combination of these methods. Avoid concentrated discharge onto fill slopes unless the fill slopes are stable and erosion-proofed.
2. Consider using broad-based drainage dips or leadoff ditches in lieu of cross drains for low volume roads. Locate these drainage features where they will not drain into wetlands, floodplains, and waters of the State.
3. Avoid use of outside road berms unless designed to protect road fills from runoff. If road berms are used, breach to accommodate drainage where fill slopes are stable.
4. Design roads crossing low-lying areas so that water does not pond on the upslope side of the road. Provide cross drains at short intervals to ensure free drainage.
5. Divert road and landing runoff water away from headwalls, slide areas, high landslide hazard locations, or steep erodible fill slopes.

Cross Drain BMPs:

1. Locate cross drains to prevent or minimize runoff and sediment conveyance to waters of the State. Implement sediment reduction techniques such as settling basins, brush filters, sediment fences, and check dams to prevent or minimize sediment conveyance. Locate cross drains to route ditch flow onto vegetated and undisturbed slopes.
2. Choose cross drain culvert diameter and type according to predicted ditch flow, debris and bedload passage expected from the ditch. Minimum diameter is 18.”
3. Locate surface water drainage features (e.g., cross drain culverts, rolling dips, and water bars) where water flow will be released on convex slopes or other stable and non-erosive areas that will absorb road drainage and prevent sediment flows from reaching wetlands, floodplains, and waters of the State. Where possible, locate surface water drainage structures above road segments with steeper downhill grade. Locate cross drains at least 50 feet from the nearest stream crossing and allow for a sufficient non-compacted soil and vegetative filter.
4. Armor surface drainage structures (e.g., broad based dips, and leadoff ditches) to maintain functionality in areas of erosive and low-strength soils.
5. Discharge cross drain culverts at ground level on non-erodible material. Install downspout structures or energy dissipaters at cross drain outlets or drivable dips

where alternatives to discharging water onto loose material, erodible soils, fills, or steep slopes are not available.

6. Cut protruding ‘shotgun’ culverts at the fill surface or existing ground. Install downspout or energy dissipaters to prevent erosion.
7. Skew cross drain culverts 45–60 degrees from the ditch line and provide pipe gradient slightly greater than ditch gradient to reduce erosion at cross drain inlet.
8. Provide for unobstructed flow at culvert inlets and within ditch lines during and upon completion of road construction prior to the wet season.
9. Install critical dips down-grade of cross drain culverts to accommodate overtopping flows. Where road grades are too adverse for critical dip construction, provide for appropriate routing of overtopping flows across road tread or into adjacent, suitably-sized drainage structure.

General Erosion Control BMPs:

1. Where sediment could be transported to waters of the State, place sediment-trapping materials or structures such as straw bales, jute netting, or sediment basins at the base of newly constructed fill or side slopes. Keep materials away from culvert inlets or outlets.
2. Suspend ground-disturbing activity if projected forecasted rain will saturate soils to the extent that there is potential for movement of sediment from the road to wetlands, floodplains, and waters of the State. Cover or temporarily stabilize exposed soils during work suspension. Upon completion of ground-disturbing activities, immediately stabilize fill material over stream crossing structures. Measures could include but not limited to erosion control blankets and mats, soil binders, soil tackifiers, or placement of slash.